

Samuel Koovely

Building Y27, Office G25
Winterthurerstrasse 190, 8057 Zürich
samuel.koovely@math.uzh.ch

EXPERIENCE	<i>Committee member for professorial appointment</i>	Sep. 2024 - Ongoing
	Department of Mathematical Modeling and Machine Learning, UZH, Switzerland	
	<i>Committee member for professorial appointment</i>	Jun. 2023 - Dec. 2023
	Institute for Mathematics, UZH, Switzerland	
	<i>Teaching Assistant</i>	Sep. 2022 - Ongoing
	University of Zurich, Switzerland	
	• Organizer of <i>Introduction to Machine Learning</i> (English)	Spring Term 2025
	• Teaching Assistant of <i>Complex Networks Theory and Applications</i> (English)	Fall Term 2024
	• Head Teaching Assistant of <i>Complex Networks Theory and Applications</i> (English)	Fall Term 2022/2023
	• Organizer of <i>Seminar on Markov Models</i> (English)	Spring Term 2024
EDUCATION	<i>Project Intern and Trainee</i>	Nov. 2020 - Nov. 2021
	IBM Research Zurich, Switzerland	
	• Project intern for my MSc Thesis [2]	
	• Further developed the model introduced in [2] as a post-graduate trainee.	
	<i>Teaching Assistant</i>	Sep. 2017 - Aug. 2019
	ETH Zurich, Switzerland	
	• Assistant of <i>Mathematik IV: Statistik</i> (German)	Spring Term 2019
	• Assistant of <i>Stochastics (Probability and Statistics)</i> (German)	Fall Term 2018
	• Assistant of <i>Probability and Statistics</i> (English)	Fall Term 2017
	<i>PhD studies in Mathematics</i>	Aug. 2022 - ongoing
SKILLS	University of Zurich, Institute of Mathematics, Institute of Mathematical Modeling and Machine Learning	
	<i>BSc and MSc studies in Mathematics</i>	Sep. 2015 - Jul. 2021
	ETH Zurich, Department of Mathematics	
	Languages: Italian (Native tongue), English (proficient), French (good), German (good)	
	Programming Skills: Python (good), R (intermediate), Matlab (intermediate), C++ (basic), Wolfram Language (Basic)	
	<i>Entropy-Informed Structural Changes Detection</i> , NetSci 2025	Jun. 2025
	<i>What is... Free Convolution?</i> , ZGSM Graduate Colloquium, ETH/UZH	Mai 2025
	<i>Introduction to Machine Learning</i> , UZH (organizer)	Spring 2025
	<i>Diffusion on Graphs</i> , DM3L Inaugural Symposium 24, UZH	Sep. 2024
	<i>Information-theoretic framework for the dynamics of temporal networks</i> ,	
SEMINARS & CONFERENCES		

	NetSci 2024	Jun. 2024
	<i>Seminar on Markov Models</i> , UZH (organizer)	Spring 2024
	<i>Random Walks and Community Detection</i> , Complex Networks: Theory, Methods, and Applications - 7th ed.	May 2023
	<i>The art of statistics</i> , ETH	Dec. 2019
	<i>Theory and applications of Machine Learning</i> , ETH	Nov. 2019
	<i>Representation theory: groups, algebras and quivers</i> , ETH	Oct. 2018
	<i>Vector Bundles in algebraic topology</i> , ETH	Mar. 2018
SERVICES TO COMMUNITY	<i>Oragnization</i> : ZGSM Graduate Colloquium, ETH/UZH	Jul. 2025-ongoing
	<i>Reviewer for Journals</i> : Physical Review E (2025)	
	<i>Reviewer for Conferences</i> : NetSci (2025)	
	<i>Examiner of candidates high-school teacher in mathematics</i> , Kantonsschule Rämibühl MNG	Oct. 2024
	<i>Committee member for professorial appointments</i> , UZH	2023-2025
PREPRINTS & PUBLICATIONS	[1] Alexandre Bovet and Samuel Koovely. “Entropy-Informed Temporal Community Detection”. In: <i>Manuscript in preparation</i> (2025).	
THESES & PROJECTS	[2] Samuel Koovely. “A mathematical framework for COMIC-Tree: an undirected graphical model for T-cell receptors specificity”. MSc Thesis. 2021.	
	[3] Samuel Koovely. “Overview and empirical evaluation of some variations of the PC-algorithm”. Semester Research Project. 2020.	
	[4] Samuel Koovely. “Introduction to Riemann surfaces and covering spaces”. BSc Thesis. 2019.	